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Indonesian national | Available for a full-time internship placement | Fully mobile within Indonesia

PROFILE

Electrical engineering undergraduate whose core discipline is measurement: designing the instrument, calibrating it against a reference, and stating the uncertainty on the number it produces. I currently apply that discipline to a live production data pipeline, where I define quality metrics, instrument each stage of the process to measure them, and validate every detector against deliberately injected faults before claiming it works. That method has already exposed two defects that reviewing the process alone would not have found. Applying to the Toyota Internship Program in engineering, quality, or production control.

EDUCATION

Institut Teknologi Bandung (ITB) | Bandung, Indonesia
Bachelor of Engineering (S1), Electrical Engineering 2023 to [graduation year]

Academic standing: cumulative GPA 2.49 / 4.00. Most recent semester 3.07, the strongest of my degree so far. My early semesters were poor and I own that. The upward trend is deliberate and I am working to raise the cumulative figure before graduation.

Relevant coursework: Instrumentation Systems, Control Systems, Electronics I and II, Digital Signal Processing, Electrical Power Systems, Embedded Systems, Microprocessor Systems, Probability and Statistics.

EXPERIENCE

Data Scientist Intern | PT Tricada Intronik (Tritronik), Bandung
2026 to present

Built a data quality monitoring framework for a live telecommunications processing pipeline. The system answers a question that ordinary status monitoring cannot: not whether the line is running, but whether what comes off it is correct.

- Defined a four-dimension quality standard (completeness, timeliness, freshness, consistency) with explicit metrics, formulas and inspection points at each stage of the process, so that improvement could be measured rather than asserted.
- Instrumented every stage in Python against a live Apache Kafka broker and PostgreSQL database, measuring throughput counts, processing latency and stage-to-stage reconciliation.
- Built predictive alerting using rate-of-change and time-to-breach estimation, so a developing problem is flagged before it becomes a defect rather than after.
- Validated every detector against datasets with deliberately injected faults of known type, magnitude and timing, treating detection capability as a measured result rather than an assumption.
- Found and corrected a defect in which a source that stopped producing generated no data to inspect, so the monitor reported normal while structurally blind to the stoppage. Detection precision rose from 71% to 100% with no threshold change.
- Measured a 29x reduction in query time from indexing on a 4-million-row table, and made index specification a deployment requirement rather than a recommendation.

Why this matters for manufacturing: a monitoring system that hides its own abnormality is worse than no monitoring, because it manufactures false confidence. Finding that defect required going to the process and testing it against faults whose answers I already knew.

Project Manager and Coordinator | Himpunan Mahasiswa Elektroteknik (HME) ITB
2025

- Ran the full production lifecycle of a large-scale merchandise programme against a fixed deadline: specification, vendor negotiation, design iteration, quality review and on-time delivery.
- Coordinated a cross-functional team without direct line authority, consolidating conflicting stakeholder feedback into decisions across repeated review cycles.
- Set pricing by benchmarking production cost against market rates, protecting margin while keeping the product accessible.

Also with HME ITB: Staff of Fundraising, Masa Bina Cinta orientation programme (2025); Event Organizer Team, university graduation event (2024).

SELECTED PROJECTS

PID-Controlled Magnetic Levitation: Stabilising an Open-Loop-Unstable Plant | Control Systems, ITB

- Derived the nonlinear model from first principles, then linearised about equilibrium into a 3-state model with a right-half-plane pole at $s = +40.41$, and confirmed instability three independent ways rather than trusting one method.

- The first controller overshoot by 239.23%. Detuning would have produced an acceptable-looking response, but it would have treated the symptom: I traced the cause to controller-zero excitation by the discontinuous step input, a structural problem and not a tuning error.
- Designed a pre-filter cancelling the PID zeros ahead of the summing junction. Final result: 4.409% overshoot, 0.6287 s settling, zero steady-state error, original design fully preserved.

Why this matters for manufacturing: *the countermeasure that makes the symptom disappear is rarely the one that removes the cause, and only the second kind survives.*

Precision NTC Thermistor Instrument | Instrumentation Systems, ITB

- Engineered the complete signal-conditioning front-end, a Wheatstone bridge feeding a high-CMRR instrumentation amplifier, achieving sub-0.1 C resolution while rejecting common-mode noise and supply interference.
- Applied Steinhart-Hart linearisation, validated the instrument through multi-point calibration against certified references, and produced a GUM-compliant uncertainty budget stating the confidence interval on every reading.

Why this matters for manufacturing: *an inspection gauge is only as good as its calibration traceability. A measurement without a stated uncertainty cannot support a pass-fail decision.*

70 kV Mesh Power System: Emergency Operation and N-2 Redundancy | EL3017, ITB, team of 5

- Simulated every contingency across peak and night loading, confirming 0.969 to 1.010 pu bus voltages in all cases and validating uninterrupted supply under simultaneous dual-feeder failure. Directly relevant to facility reliability and uninterrupted line operation.

Nephro-Sync: Wearable Biosensor and Telepharmacy Ecosystem | EPIDEMIC 2026 National Scientific Essay Competition

- Co-designed a conceptual wearable sensing system; I designed the sensing, control, edge and power layers. Identified thermal drift as the dominant error source rather than the headline feature, and specified a PID-controlled heater holding the sensor microclimate at 37 C so an alert reflects real physiology and not instrument drift. Design is unvalidated; all figures are design targets.

SATU-TB: Edge-AI Tuberculosis Screening Device | National Essay Competition, 6th of 577 teams

- Co-designed a low-cost dual-modality sensing device on a Raspberry Pi and Arduino architecture, moving feature extraction on-device to cut RAM by 50% and energy by a factor of 20, enabling operation where there is no connectivity.

TECHNICAL SKILLS

Measurement and quality: Sensor signal conditioning, multi-point calibration against certified references, GUM-compliant uncertainty budgeting, sensor linearisation, fault-injection validation, root-cause investigation.

Control and process: PID design and tuning, pole placement, state-space modelling, closed-loop stability analysis (root locus, Bode, Nyquist), MATLAB and Simulink.

Electronics and embedded: Analog front-end design, SPICE transient verification, DC-DC converters and ripple filtering, bare-metal C and C++ firmware, I2C and SPI, ESP32, Arduino, Raspberry Pi, LTspice.

Data: Python, pandas, SQL, PostgreSQL, Apache Kafka, Docker, Git, Linux.

Languages: Indonesian (native), English (fluent; EF SET C1 Advanced, 70/100).

HONOURS AND CERTIFICATIONS

- 6th place of 577 teams, national scientific essay competition (SATU-TB). Co-author, EPIDEMIC 2026 national scientific essay competition (Nephro-Sync).
- English proficiency: EF SET English Certificate, 70/100, C1 Advanced (EF SET, 2025).
- Test and measurement: Signal Generator Fundamentals; Power Supplies Fundamentals (Rohde und Schwarz, 2026).
- Systems and embedded: IoT Systems and Industrial Automation (LearnQuest, 2026); Interfacing with the Arduino (UC Irvine, 2026); Unix and Linux fundamentals, five-course series (EDUCBA, 2026).